Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

HHS Public Access
Author manuscript
Cell. Author manuscript; available in PMC 2016 July 16.

Published in final edited form as:
Cell. 2015 July 16; 162(2): 375–390. doi:10.1016/j.cell.2015.06.034.

FOXG1-dependent dysregulation of GABA/glutamate neuron differentiation in autism spectrum disorders

Jessica Mariani[1,2,*], **Gianfilippo Coppola**[1,2,*], **Ping Zhang**[3], **Alexej Abyzov**[1,4,&], **Lauren Provini**[1,2], **Livia Tomasini**[1,2], **Mariangela Amenduni**[1,2], **Anna Szekely**[1,5], **Dean Palejev**[1,2,¶], **Michael Wilson**[1,2], **Mark Gerstein**[1,4,6,7], **Elena Grigorenko**[1,2], **Katarzyna Chawarska**[1,2], **Kevin Pelphrey**[1,2], **James Howe**[3], and **Flora M. Vaccarino**[1,2,8]

[1]Program in Neurodevelopment and Regeneration, Yale University, New Haven, Connecticut 06520, USA

[2]Child Study Center, Yale University, New Haven, Connecticut 06520, USA

[3]Department of Pharmacology, Yale University, New Haven, Connecticut 06520, USA

[4]Department of Molecular Biophysics and Biochemistry, Yale University, New Haven, Connecticut 06520, USA

[5]Department of Genetics, Yale University, New Haven, Connecticut 06520, USA

[6]Program in Computation Biology and Bioinformatics, Yale University, New Haven, Connecticut 06520, USA

[7]Department of Computer Science, Yale University, New Haven, Connecticut 06520, USA

[8]Department of Neurobiology, Yale University, New Haven, Connecticut 06520, USA

SUMMARY

Autism spectrum disorder (ASD) is a disorder of brain development. Most cases lack a clear etiology or genetic basis, and the difficulty of reenacting human brain development has precluded understanding of ASD pathophysiology. Here we use three-dimensional neural cultures (organoids) derived from induced pluripotent stem cells (iPSCs) to investigate

Correspondence and requests for materials should be addressed to: flora.vaccarino@yale.edu.
*These authors contributed equally to this work
&Present address: Department of Health Sciences Research, Mayo Clinic, Rochester, MN 55905
¶Present address: Bulgarian Academy of Sciences, Institute of Mathematics and Informatics, Sofia 1113, Bulgaria

Supplementary Information is linked to the online version of the paper.

Author contribution

The authors contributed this study at different levels, as described in the following. Study conception and design: F.M.V., J.M., G.C., J.H. Family selection and characterization: K.C., K.P. Skin biopsy: A.S. iPSC generation and characterization: J.M., L.T., L.P., M.A. Neuronal differentiation: J.M., L.T, L.P. Electrophysiology: P.Z., J.H. RNAi: J.M. Processing and analysis of RNAseq data: G.C., D.P. Processing and analysis of DNAseq data, CNV and SNV discovery: A.A., M.W. qPCR validation: J.M. Western blotting: J.M. Human subjects: K.C. and K.P. Coordination of analyses: F.M.V. Display item preparation: J.M., G.C., F.M.V., L.T., J.H. Writing manuscript: J.M., G.C., F.M.V., J.H. All authors participated in discussion of results and manuscript editing.

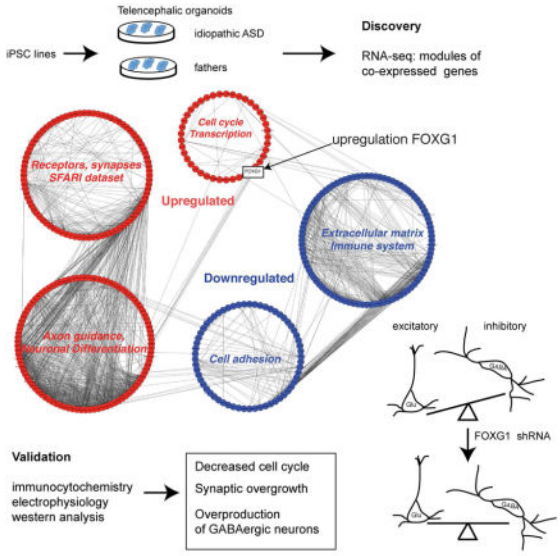
Competing Interest: The Authors declare no competing interest

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

neurodevelopmental alterations in individuals with severe idiopathic ASD. While no known underlying genomic mutation could be identified, transcriptome and gene network analyses revealed upregulation of genes involved in cell proliferation, neuronal differentiation, and synaptic assembly. ASD-derived organoids exhibit an accelerated cell cycle and overproduction of GABAergic inhibitory neurons. Using RNA interference, we show that overexpression of the transcription factor FOXG1 is responsible for the overproduction of GABAergic neurons. Altered expression of gene network modules and FOXG1 are positively correlated with symptom severity. Our data suggest that a shift towards GABAergic neuron fate caused by FOXG1 is a developmental precursor of ASD.

Graphical Abstract



INTRODUCTION

Rare penetrant mutations, common genetic variants and environmental factors are known to contribute risk to ASD, yet, about 80% of the cases have no clear etiology and no pathogenetic model. A large number of rare mutations have been identified in the context of syndromic and non-syndromic ASD and have been modeled in various organisms. However, these mutations are extremely heterogeneous, each accounting for less than 1–2% of cases. Furthermore, in no instance have they been shown to be sufficient to cause ASD; rather, they interact with other inherited and non-inherited risk factors. Some mutations involve synapse-associated molecules (Sudhof, 2008; Zoghbi and Bear, 2012) and have led to the widespread notion that alterations in the assembly of synaptic connections are key in the pathophysiology of ASD. Others have formulated the hypothesis that there is an excitatory/inhibitory neuron imbalance in the disorder (Casanova et al., 2003; Rubenstein, 2010).

The applicability of these pathogenetic mechanisms to the great majority of ASD cases remains unknown. Furthermore, the heterogeneity of phenotypes found when modeling these mutations in animals and the inherent difficulty in creating behavioral phenotypes of

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

ASD in rodents have complicated the construction of credible animal models of ASD. It is possible that the heterogeneity of rare mutations found in ASD, as currently conceptualized, denies a unified understanding of the pathophysiology of the disorder. However, emerging evidence suggests that current genomic data, when considered in the framework of gene network analyses, point to a common pathophysiological substrate in ASD rooted in the embryonic development of the cerebral cortex (Parikshak et al., 2013; Willsey et al., 2013).

Here, we have taken the approach of directly modeling early cortical development in probands with idiopathic ASD. We focused on individuals with increased head/brain size (macrocephaly), as this is one of the most consistently replicated ASD phenotypes (Courchesne et al., 2001) and confers poorer clinical outcomes ASD (Chaste et al., 2013; Chawarska et al., 2011; Lainhart et al., 2006). Using induced pluripotent stem cells (iPSCs) obtained from affected families, we have produced telencephalic organoids that recapitulate transcriptional programs present in mid-fetal human cortical development. Transcriptome and cellular phenotype analyses in this model identified unexpected differences in cell cycle time and synaptic growth, as well as an imbalance in GABA/glutamate neuronal differentiation in patients as compared to their unaffected family members.

RESULTS

Genomic analysis of iPSC derived from patients with idiopathic ASD

To analyze neurodevelopmental aspects of idiopathic ASD, we generated iPSC lines from members of four families that each included an ASD proband with increased head circumference (HC) and one to three unaffected, first-degree family members (see Figure S1A for family structure and Table S1 for data collection and participants details).

Whole genome sequencing data was obtained on all fibroblasts and iPSCs for members of all families in this study. Data were analyzed with CNVnator (Abyzov et al., 2011) for copy number variations (CNVs) discovery and with a GATK-based pipeline for single nucleotide variation (SNV) discovery (see Supplemental Experimental Procedures). For three families with both parents participating in the study (03, 07, and 1123), we predicted *de novo* CNVs and SNVs. Of the putative *de novo* CNVs, only one 4.8 kb deletion (chr14:39987476-39992327) found in the 1123 proband could be validated by qPCR. For this event, qPCR validation showed roughly 30% copy-number decrease in the proband relative to either of the parent, suggesting that the deletion is possibly a somatic mosaic variant. This deletion did not overlap any known genes. Proband 07-03 carries a previously uncharacterized deletion involving exon 2 in the PTEN gene (chr10:89,641,498-89,658,394), which was also found in his unaffected father. Proband 1120 and 03 did not carry any deleterious CNVs that could potentially be pathogenic. On average, each proband had ~112,000 rare SNVs not previously detected by the 1000 Genomes Project. This is also likely to include most of *de novo* SNVs. By intersecting rare SNVs in probands with lists of genes whose disruption has been previously linked to ASD (Willsey et al., 2013) as well as the SFARI syndromic genes dataset (https://gene.sfari.org/autdb/Welcome.do), we found no rare SNVs that cause a known deleterious loss of function of an ASD gene. In summary, we found no *de novo* CNVs and/or no rare SNVs in probands

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

that cause a known deleterious loss of function in the protein coding sequence of a gene previously involved in rare cases of syndromic or non-syndromic ASD.

Patient iPSC-derived telencephalic organoids show grossly normal organization and neuronal excitability

For the four families, we differentiated two to three iPSC lines per person into organoids using a modification of our free-floating tridimensional (3D) culture method (Mariani et al., 2012) (see schematic outline in Figure S3A). Similarly to our previously published preparation (Abyzov et al., 2012; Mariani et al., 2012) and to other protocols described in the literature (Eiraku et al., 2008; Lancaster et al., 2013) we obtained organoids characterized by autonomously organized layers of radial glia, intermediate progenitors and neurons (Figure 1; Figure S3C, D).

After 11 days of terminal differentiation (TD11), the organoids were composed of polarized, proliferating progenitors expressing the radial glial cell markers BLBP, NESTIN, PAX6, BRN2, SOX1/2. The radial glial cells underwent mitoses on the apical (luminal) side, whereas neuronal precursor cells expressing the immature neuronal proteins DCX and TUBB3, and more mature $NeuN^+$ neurons after 31 days of terminal differentiation (TD31), accumulated on the basal side of the layer of radial glial cells (Figure 1A; Figure S3C, D). Both ASD-derived and unaffected family member-derived iPSCs had an equivalent potential to generate neuronal cells (Figure 1A, B; Figure S3C, D). We observed no significant change in either perimeter or area of the organoids between control and affected groups, except for a transient increase in ASD-derived organoid size at day 11 (Figure S3B). This is most likely due to intrinsic factors, since the starting number of iPS cells seeded per aggregate was the same in all experiments and the size of EBs at the beginning of the differentiation was not different between patients and controls (see Detailed Experimental Procedures).

To compare the electrical excitability of iPSC-derived neurons from probands with those from familial controls, we made whole-cell patch-clamp recordings from neurons in dissociated cultures or at the edge of organoids. All neurons examined (n=99) expressed voltage-activated sodium and potassium currents that were of similar amplitude in control and proband neurons from two different families (Figure S3E–G). Because we observed that iPSC differentiated using dissociated monolayers were not able to robustly generate ventral telencephalic neurons and express virtually no $GAD1^+$ cells, we limited our experiments to organoid preparations. In recordings from neurons in the organoids, the voltage-activated currents supported action potential firing in 18/18 control neurons and 12/14 proband neurons, with thresholds of −42.5 ± 0.8 mV and −37.9 ± 2.1 mV (controls, probands; p = 0.06) and action potential overshoots of 30 to 65 mV. Most neurons fired only a single action potential, but some fired multiple spikes (Figure 1C). In total, the electrophysiological data indicate that the cells studied here display voltage-gated channels similar to those in central neurons. In addition to these signatures of electrical excitability, we also recorded spontaneous synaptic currents in some neurons (Figure 1D–F). In three control neurons (of 20 tested), the fast rise (0.5–0.7 ms) and decay (1.8–2.4 ms) of these currents and their reversal near 0 mV strongly suggested that they were AMPA-receptor EPSCs (Figure 1D–

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

F). In three other neurons (2 of 17 from patients, 1 additional control) we saw occasional synaptic currents that were larger (40 to 325 pA, at −70 mV) and which decayed with time constants of 7 to 10 ms. Although the percentage of neurons showing synaptic currents was low, the results clearly demonstrate that neurons in the organoids form functional synaptic connections.

Transcriptome analysis of organoids derived from ASD individuals

To assess both the region specificity and maturity of our organoid preparation in a comprehensive fashion, we analyzed the global transcriptome of organoids by RNA-seq at each of three time points (rosette stage, TD11, and TD31, two to three iPSC lines per individual, total 45 samples). The organoid's transcriptomes were compared with BrainSpan, the largest dataset of postmortem human brain transcriptomes from embryonic age to adulthood (Kang et al., 2011). This comparison indicated that our preparation best reflected the transcriptome of the human brain during early fetal human development (9 weeks post conception), with TD31 cells displaying significant correlations also with second trimester human brain samples, up to 13–16 weeks post-conception (Figure 1G). With regards to regional specification, the transcriptome of iPSC-derived organoids was most similar to the human dorsal telencephalon (cerebral cortex and hippocampus), particularly to dorsolateral, medial, and orbitofrontal cortical areas, with smaller but significant homology to the cerebellar anlage and the amygdala (Figure 1H, I).

Next, we compared the transcriptomes of the four probands to those of the unaffected family members (two to three iPSC clones per person) at two time points, TD11 and TD31. Differential gene expression (DGE) between the probands and the respective fathers, used as sex-matched normal controls, identified 1062 differentially expressed genes (DEGs) at TD11 and 2203 DEGs at TD31 (see Table S2A), hinting at a possibly divergent developmental trajectory between controls and probands. Validation by qPCR of a subset of the DEGs identified by RNAseq revealed a 0.98 correlation coefficient between log2 fold changes from the two techniques and 100% concordance in direction of change (see Table S2B). The individual-to-individual (biological) variability, as modeled in RNA-seq DGE analyses, is clearly narrow (Table S2). The iPSC line-to-line variability is also quite low, as shown by the boxplots of correlation coefficients within each individual (intra) and across individuals (inter) (Figure S4). In fact, the variability between lines from the same individual is lower than the variability between lines across individuals (Figure S4A–C). This reproducibility is likely due to the robustness of our telencephalic organoid preparation, which selects for forebrain progenitors while allowing spontaneous 3-D organization.

We then asked whether, at a system level, the DEGs underline coherent alterations in groups of co-expressed genes. Using weighted gene co-expression network analysis, WGCNA (Langfelder and Horvath, 2008), we identified 24 modules of co-expressed genes across probands and controls at TD11 and TD31, all of which survived permutation analysis (Table S3). We estimated the modules' eigengenes (i.e. the first principal component of the module's genes expression profiles) and assessed their changes over time and across diagnosis (see Figure 2A). The *yellow* and *green* modules (annotated by "vascular development" and "lipid metabolism" gene ontology (GO) categories, respectively) were

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

enriched in downregulated DEGs (Table S4), and their eigengenes were consistently downregulated across time points (Figure 2A). The *blue* and *magenta* modules (annotated by "neuronal differentiation" and "regulation of transcription" GO terms, respectively) were enriched in upregulated genes at both TD11 and TD31 (Table S4), and their eigengenes were consistently upregulated across time points, in keeping with their developmental annotation. The *brown* and *tan* modules (annotated by "synaptic transmission" and "gated channel activity" GO terms, respectively), however, were enriched in upregulated genes only at TD31 (Table S4), and consistently, their eigengenes were upregulated more at TD31 than at TD11, in line with their synaptic functional annotation (Figure 2A). Hierarchical clustering of module eigengenes also showed a tighter correlation among the *green* and *yellow*, the *blue* and *magenta*, and the *brown* and *tan* modules (Figure 2B), suggesting similar functions and/or tighter regulatory interactions. In sum, probands were characterized on the transcriptome level by a downregulation of non-neuronal and a corresponding upregulation of neuronal transcript modules.

We further refined our understanding of the functional annotations of the upregulated *magenta*, *blue*, *brown* and *tan* "neuronal" modules by investigating their canonical pathways annotations. The *magenta* module was significantly enriched in transcription- and cell cycle-related canonical pathways (Tables S5, S6). Indeed, among the significantly upregulated genes in this module are a number of transcription factors (TFs) crucial for acquisition of neural cell fates and precursor cell proliferation in the telencephalon, including DLX6-AS1, FOXG1, EOMES, POU3F3/BRN1, SOX3, SOX5, GSX2, ETV1, DLX1, DLX6, E2F2, and SYNE2 (Tables S4, S5, S6). Many of these TFs are also hub genes (Figure 2C; Table S7). Overall, genes in the *magenta* module function in the transcriptional regulation of cell fate and cell proliferation in the forebrain.

For the *blue* "neuronal differentiation" module, canonical pathway annotation was enriched in axon guidance and generic transcription pathways (Tables S5, S6). The axon guidance genes, all upregulated in probands, included members of the neural cell adhesion family (NCAM1, NRCAM, L1CAM, NFASC), semaphorin (PLXNC1, PLXNB3), netrin (DCC, UNC5A), Rho GTPases (PAK3, PAK7), CDK5R1/P35, and DCX. Many of these molecules were among the top 100 hub genes, i.e., CDK5R1, DPYSL5, DCX, DPF1, APC2 (Figure 2C; Table S7). Hence, genes in the *blue* module are involved in cytoskeletal regulation of various cellular functions, including neurite outgrowth, axon guidance, cell proliferation, migration, and survival.

For the *brown* and the *tan* modules, the GO and top pathways annotation were mostly related to synaptic functions, ion channels, and ligand-receptor interactions (Tables S5, S6), which is in harmony with their more significant eigengene upregulation in probands at the later time point (Figure 2A). The *brown* module displayed the strongest enrichment with respect to the SFARI autism gene dataset, a collection of genetic information that includes data from linkage and association studies, cytogenetic abnormalities, and specific mutations associated with ASD (https://gene.sfari.org) (SFARI category S to 4)(Abrahams et al., 2013) (p-value = 6.51E-5). Among the top 100 hub genes of the *brown* module, 64 were upregulated at TD31, including molecules involved in synaptic assembly, ion transport, and post-synaptic signaling (i.e., NRXN1, NRXN2, SLITRK1, CAMK2B, CAMK1D, NRSN1,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

SYT13, GRIN1, SCN2A) (Table S7). Among the hub genes, NRXN1, SCN2A, and TSPAN7 were both significantly upregulated and overlapping with genes in the SFARI autism database (Figure 2C). The *tan* module was characterized by an upregulation at TD31 of transcripts for several potassium channels and for key components of GABAergic neurons, including the GABA synthetic enzyme GAD1 and three GABA receptor subunits, most of which were in the *tan* module's 100 hub genes (Table S7).

Altered neural proliferation and differentiation in organoids from ASD patients

In summary, the signatures that emerged from our transcriptome analyses as significantly perturbed in probands were transcriptional regulation of cell proliferation/cell fate, neuronal differentiation/process outgrowth, and synaptic transmission. To functionally validate these signatures, we performed morphometric cellular analyses and immunostaining for cell fate markers. We first studied the dynamics of the cell cycle in undifferentiated iPSCs and neuronal progenitors (TD11) by BrdU incorporation (Figure 3A, B). These experiments revealed a significant decrease in cell cycle length in ASD-derived iPSCs (Figure 3A). We saw a similarly strong trend in early neuronal progenitors cultured as monolayers (Figure 3B). However, when we estimated the proportion of proliferating cells in more mature organoids (TD31), there was no significant difference in proliferation between ASD- and control-derived organoids (Figure 3C, D). Taken together, these results suggest that a decrease in cell cycle length could be an early event that is present at the iPSC undifferentiated stage and during the early stages of neuronal differentiation, which is consistent with the transient increase in organoid size at TD11 (Figure S3B).

We then assessed neuronal maturation and synaptic formation in organoids at TD31. Quantification of microtubule-associated protein 2 (MAP2) showed a significant increase in its density in ASD-derived neurons (Figure 3E, F). Moreover, synapse number quantification revealed a significant increase in Synapsin I-immunoreactive $(SynI)^+$ puncta in ASD-derived neurons, suggesting increased neuronal maturation and synaptic overgrowth (Figure 3E, G). This is in agreement with the upregulated expression of the *blue* and *brown* modules (Figure 2A), whose signatures suggested accelerated or increased neuronal differentiation and synaptic connections in probands. Furthermore, quantification of the inhibitory VGAT (vesicular GABA transporter) and excitatory VGLUT1 (vesicular glutamate transpoter-1) immunoreactivity revealed a significant increase in VGAT-immunoreactive puncta (Figure 3H, I) and no significant changes in VGLUT1 puncta (Figure 3J), suggesting an increase in the number of inhibitory synapses in ASD-derived neurons.

Next, we directly tested whether there was any bias for differentiation into specific neuronal subtypes. In this analysis, we used as markers TFs which control fate choice, cell proliferation, and neuronal differentiation during normal telencephalic development, many of which were members of the *magenta* and *blue* modules (Figure 2C; Tables S5, S6). We found that the proportions of cortical excitatory neuron precursors of the subventricular zone expressing EOMES/TBR2, of layer 6 neurons expressing TBR1, and of early-born layer 5 CTIP2 neurons (also known as BCL11B) were not significantly different in ASD and control organoids at TD31 (Figure 4A–F), although some of these cortical excitatory neuron

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

markers (such as TBR1, TBR2, CTIP2 and SOX5) were upregulated in proband-derived organoids at the mRNA level.

We next investigated determinants of GABAergic inhibitory neuronal fates using DLX1-2 (two TFs which are among the earliest determinants of GABAergic fate in telencephalic precursors cells and upregulated members of the *magenta* module) and GAD1/GAD67 (the GABA synthetic enzyme, an upregulated member of the *tan* module). The expression of these GABAergic markers was increased significantly in organoids derived from ASD individuals compared to those from unaffected family members. This increase was strong at TD31 and was already detectable at TD11 (Figure 4G–J). The increase in DLX1/2 and GAD1/GAD67 was consistent in probands irrespectively of iPSC line, or individuals within different families (Figure S4G–L). Also, the increases in DLX1/2 and GAD1 in probands with respect to their fathers were reproducible across independent differentiation experiments (Figure S4M).

GABA precursors arose in a segregated fashion in an area of the organoids and did not colocalize with $TBR1^+$ excitatory neuron precursors (Figure S5). The number of cells immunoreactive for ASCL1/MASH1 and NKX2.1 (two TFs expressed by GABAergic progenitor cells) and the neurotransmitter GABA was also increased in ASD-derived organoids (Figure S5). Moreover, western blot analyses further confirmed increased protein expression of GAD1/2 in ASD organoids (Figure S5Q, R). Along with the upregulation of GSX2, ASCL1, DLX1, DLX2, DLX5, DLX6 and DLX6-AS1, which is the top upregulated gene at TD11 (Table S2) and the increase in GABA transporter immunostaining (Figure 3H, I), these cellular analyses strongly suggest an overproduction of progenitors and neurons of the GABAergic lineage as well as an altered balance between the number of excitatory and inhibitory neurons in organoids derived from probands (Figure 4K).

We also found evidence for increased expression of GABAergic phenotypes electrophysiologically. Although the size of sodium currents in control- and proband-derived neurons was similar (Figure S3), there was substantial cell-to-cell variation in the voltage dependence of activation and inactivation, suggesting that different neurons express different proportions of brain sodium channel isoforms ($Na_v1.1$, $Na_v1.2$, $Na_v1.3$, $Na_v1.6$). Figure 5 shows results for steady-state inactivation. Neurons from probands and their familial controls displayed substantial variation in the pre-test potential (V_{pre}) at which the peak sodium current was reduced to half its maximal amplitude ($E_{h1/2}$), and adequate fits to some of the data required two Hodgkin-Huxley components (Figure 5D, E). Interestingly, sodium currents in proband-derived neurons tended to inactivate at more hyperpolarized membrane potentials than the corresponding currents in control neurons (Figure 5C–E; n=7 ASD and 10 control neurons). The increased proportion of proband-derived neurons that gave $E_{h1/2}$ values in the range −72 to −65 mV (Figure 5F) is consistent with increased expression of the $Na_v1.1$ isoform in these cells. This isoform is preferentially expressed in GABAergic interneurons (Cheah et al., 2012; Han et al., 2012; Ogiwara et al., 2007; Yu et al., 2006) and the increased proportion of cells with this sodium channel phenotype is consistent with a greater proportion of GABAergic neurons in cortical organoids from probands with ASD.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

FOXG1 overexpression causes deregulated cell differentiation in ASD organoids

Our DGE results show that DLX6-AS1, TMEM132C, FOXG1, C14orf23 and KLHDC8A are consistently among the top 10 upregulated genes at both TD11 and TD31 (Table S2A). Among these genes, FOXG1, which is one of the top 100 hubs in the *magenta* module, with an 8.5- and 13-fold increase in expression at TD11 and TD31, respectively (Table S2A), is a transcription factor important for the development of the telencephalon (Hanashima et al., 2004; Martynoga et al., 2005; Xuan et al., 1995). Notably, loss-of-function mutations in FOXG1 have been found in patients with atypical Rett Syndrome (Ariani et al., 2008; Bahi-Buisson et al., 2010; Shoichet et al., 2005) and confer a small brain size (Kortum et al., 2011). As the probands under investigation have large brain size, it is possible that FOXG1 may be, at least in part, involved in the modulation of the brain size phenotype, and possibly in the social disability component of the phenotype.

We therefore tested the hypothesis that abnormally high levels of FOXG1 and its downstream genes could be responsible for the phenotypic abnormalities identified in neuronal cells of macrocephalic ASD patients. To this end, using lentiviruses carrying short hairpin RNAs (shRNAs) targeting FOXG1, we tested whether an attenuation of the FOXG1 expression level in patients' neural cells was able to revert some of the neurobiological alterations.

As proof of principle, we generated four stable iPSCs lines (three of which stably expressed different shRNAs specifically targeting FOXG1 and one expressed a non-targeting shRNA control) from a proband-derived iPSC line (07-P#9). To confirm stable downregulation of FOXG1 expression, we performed qPCR analyses at TD11 (Figure 6A). Introduction of two FOXG1-targeting shRNAs (shRNA-2 and 3) down-regulated FOXG1 mRNA expression to a level comparable to that of the unaffected family member (Figure 6A, compare bar 6 and 7 with bar 2). Immunostaining for FOXG1 confirmed that shRNA-2 and 3 were able to down-regulate its expression also at the protein level (Figure 6B–F). We next analyzed the expression of GABAergic markers after FOXG1-RNA interference (RNAi) at the transcript and protein level. At TD11, organoids derived from the iPSC lines stably expressing FOXG1 shRNA-2 and shRNA-3 (07-P#9 shRNA-2 and 3) showed downregulation of DLX1, DLX2, and GAD1 transcripts as compared to the same iPSC line expressing the shRNA control (07-P#9 shRNA-C) (Figure 6G–I). Immunostaining and stereological quantification of DLX1-2 and GAD1 positive cells showed that FOXG1 RNAi restored the normal level of GABAergic neuronal differentiation in proband-derived organoids at both TD11 and TD31 (Figure 6K–M).

These results suggest that FOXG1 is involved, at least in part, in causing the overproduction of neurons of the GABAergic lineage found in ASD-derived organoids. FOXG1 RNAi had no or minor effects on the transcript/protein expression levels of dorsal forebrain markers (such as PAX6) (Figure 6 B–F, J, L, M), or on TFs directing cortical excitatory neuron differentiation (such as TBR1) (Figure 6M).

To investigate the mechanism by which FOXG1 could affect the overproduction of GABAergic neurons, we compared cell proliferation in ASD- and control-derived organoids by BrdU incorporation with or without FOXG1 RNAi. Quantification of double-labeled

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

BrdU$^+$/Ki67$^+$ cells at TD11 revealed no general changes in proliferation between proband- and control-derived organoids (Figure 7A, B and Figure S6 A, B). However, there was a significant increase in the number of DLX2$^+$ cells that incorporated BrdU in proband-derived organoids, as well as an increased proportion of BrdU$^+$ cells that colocalized DLX1/2. Both effects were precluded by FOXG1-knockdown (Figure 7A–C). Furthermore, at TD31, proband-derived organoids showed a greatly increased proportion of DLX$^+$ and GAD1$^+$ cells that had incorporated BrdU at TD11 (Figure 7D–H), an effect that was also greatly attenuated by FOXG1 RNAi, which lowered the proportion of BrdU$^+$/DLX$^+$ and BrdU$^+$/GAD1$^+$ cells in ASD-derived organoids to levels comparable to those of the unaffected control (Figure 7D–H). Moreover, GAD1$^+$ cells at TD31 were not aberrantly entering the cell cycle, suggesting that GABAergic neurons were terminally differentiated in the organoids (**Figure S 6C**, **D**). Taken together, these data suggest that the early increase in proliferation of GABAergic neuronal progenitor cells in proband-derived organoids gave rise to an increased proportion of mature GABAergic interneurons, and that FOXG1 RNAi restored both these early and late effects to levels comparable to those found in unaffected family members. Similar experiments revealed smaller or non-significant changes in the proliferation of PAX6$^+$ and TBR1$^+$ precursor cells after FOXG1 RNAi (Figure S7), suggesting that upregulated FOXG1 expression in ASD neural cells was driving an early proliferative effect in neuronal precursor cells of the GABAergic lineage.

FOXG1 expression correlates with clinical phenotype

Previous studies suggest a strong association between increased head circumference (HC) in ASD and more severe autism symptoms and lower IQ (Chaste et al., 2013; Chawarska et al., 2011; Lainhart et al., 2006). Given the small sample size (n=4) and limited range of severity scores in the probands we were not able to evaluate these associations directly. However, in a supplementary analysis we obtained the correlations of interest after adding 5 participants who did not all meet the stringent criteria for macrocephaly that we required for deriving iPSCs (see “Participant” section in Extended Experimental Procedures) and thus display a wider spectrum of head circumference sizes. Correlation analysis in this enriched sample (n=9) indicated strong associations between HC and autism symptom severity (Spearman r = 0.79, p=0.01) (data not shown).

Next, we correlated the patients’ HC and their autism symptom severity with gene expression indices in our organoid model in the 4 families used in this study. Despite the small sample size, we found a consistent positive correlation between HC and autism symptom severity with the upregulated modules as well as a negative correlation with the downregulated modules (Figure 2D). The HC z-scores of probands displayed particularly strong correlations with all module’s eigengene and levels of FOXG1 gene expression at TD31 (Figure 2D). However, the module’s eigengenes displayed no correlation with the HC of the unaffected fathers, despite the fathers presenting the same degree of macrocephaly as the probands (see Figure 2E, F). Together, the observed patterns of correlations suggest that the upregulation in the *magenta*, *blue*, *brown* and *tan* gene network is a maladaptive trait in the probands and that it may represent a pathophysiological antecedent of symptoms.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Discussion

Our study provides a framework for studying normal human brain development and its disorders. Using iPSC-derived cortical organoids that recapitulate human first trimester telencephalic development, we performed genome-wide transcriptome analysis in four families affected by idiopathic ASD. The affected individuals do not share any obvious underlying genomic alterations, but all express a phenotypic trait that confers increased symptom severity, macrocephaly. Despite the heterogeneity in genotypes, we were able to identify perturbations in coherent programs of gene expression and associated features of altered neurodevelopment, namely upregulation of cell proliferation, unbalanced inhibitory neuron differentiation, and exuberant synaptic development.

The iPSC lines derived from ASD patients in these families show decreased cell cycle length, suggesting a generalized increase in proliferative potential. These findings are in accord with earlier hypotheses stating that abnormal control of cell proliferation and overproduction of neurons might explain the accelerated brain growth in ASD (Courchesne et al., 2011; Courchesne et al., 2007; Vaccarino et al., 2009). However, the increased proliferation is seen in early progenitors, but not at later stages of cortical neural development *in vitro*, suggesting powerful compensatory events that eventually restrict excessive production of neurons in the cortical organoids and presumably also *in vivo*.

In line with the transcriptome changes, we found biased production of neuronal subtypes from cortical neuron precursors of ASD probands, as compared with their fathers. Cortical organoids of ASD probands show, at all time points analyzed, exuberant GABAergic differentiation and no change in glutamate neuron types, which together cause an imbalance in glutamate/GABA neuron ratio. Interestingly, an unbiased stereological study in postmortem human samples showed an increase in three GABA interneuron subtypes in various subregions of the hippocampus in ASD, albeit in a small number of patients (Lawrence et al., 2010). We also found that the overproduction of GABAergic cells is attributable, at least in part, to an early increase in FOXG1 gene expression, which drives an increased proliferation and number of GABA precursor cells expressing TF driving GABAergic neuron fates, including the DLX homeobox genes (Rubenstein, 2010). FOXG1 inactivation in mice causes premature lengthening of telencephalic progenitor cell cycles and a failure to specify ventral (GABAergic) telencephalic precursors, leading to a severely hypoplastic telencephalon (Fasano et al., 2009; Martynoga et al., 2005). Hence, our data in humans are consistent with the known roles of FOXG1 in telencephalic growth as well as in the determination of GABA neuron fate.

Notably, our genomic data do not support the presence of any previously known deleterious mutation in any previously identified ASD candidate genes. Although we found no structural or sequence DNA variation in the FOXG1 locus that could explain the increase in FOXG1 gene expression (i.e., a duplication involving the FOXG1 locus or a mutation in a TF binding site in the proximal promoter region), it is possible that uncharacterized SNVs in distal regions could affect FOXG1 expression. Future research may uncover whether the gene expression dysregulation that we identify in these patients with ASD is due to novel mutations in genes that collectively affect the regulatory network of FOXG1.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

In addition to GABAergic neuron overproduction, we also find evidence of exuberant cellular overgrowth of neurites and synapses at the transcriptome and cellular levels and an increase in GABAergic synapses, consistent with the increase in GABAergic neuron number found in our preparation. This increase is consistent with increased spine densities found in the postmortem cerebral cortex of ASD patients (Hutsler and Zhang, 2010; Tang et al., 2014) as well as in the *fmr1* KO mouse, a model for Fragile X syndrome, one of the most common forms of inherited intellectual disability and ASD (Dolen et al., 2007). Consistent with the cellular phenotype, mRNA for the synaptic adhesion molecules NLGN1, NRXN1, NRXN2, and NRXN3 were all overexpressed in patient-derived organoids. In contrast to the present results, rare loss-of-function mutations in synaptic adhesion molecules (SHANK, NLGN, NRXN) suggest a deficiency in synaptic connections in some individuals with ASD. However, a gain-of-function mutation in NLGN3 (R451C) found in ASD confers an increase in GABAergic synaptic signaling (Etherton et al., 2011; Pizzarelli and Cherubini, 2013; Tabuchi et al., 2007). These apparently discrepant results may be reconciled by the suggestion that the balance, rather than absolute numbers, of glutamate and GABA neurons is important for function. Comparison of synaptic transmission and network activity between patient and control organoids would further test this hypothesis.

Thus, the early acceleration in the cell cycle, overproduction of GABA neurons and synaptic overgrowth may be antecedents of the aberrant trajectory of cortical development found in children with ASD. Neuropathological studies found an increased number of neurons (Courchesne et al., 2011), increased number of cortical minicolumns (Casanova et al., 2002) and synapses (Hutsler and Zhang, 2010; Tang et al., 2014) in unselected cases of ASD, suggesting increased production of cortical neurons. A macroscopic increase in HC, found in 15–20% of ASD cases, confers a poorer outcome in ASD (Chaste et al., 2013; Chawarska et al., 2011; Lainhart et al., 2006), suggesting that they represent the extreme of a continuum. Support for this hypothesis comes from the finding that in our patient cohort there is a correlation between the extent of change in gene expression in organoids and degree of patient's macrocephaly and symptom severity. The finding that altered gene expression does not correlate with HC in the unaffected fathers, despite the fathers having the same degree of macrocephaly, suggests that the macrocephaly in the fathers and in the probands are not equivalent, and that additional genetic factors may control the head size phenotype in ASD. Some studies also detected a different pattern of brain enlargement between patients with ASD and controls (Piven et al., 1996). Although the small number of participants precludes a definitive conclusion, our data indicate that the alterations in the dynamics of brain growth and differentiation discovered using the organoid model represent a core feature of the disorder, rather than an incidental finding, and that dysregulated gene expression in iPSC-derived organoids, and FOXG1 in particular, could be used as potential biomarkers of severe ASD.

Reinforcing the likely maladaptive function of altered levels of FOXG1, deletions and missense mutations in this gene have been associated with an atypical Rett Syndrome and small brain size (Ariani et al., 2008; Bahi-Buisson et al., 2010; Mencarelli et al., 2010). This is interesting, as it suggests that deviations in FOXG1 levels during brain development, in

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

excess and defects, cause opposite modulation in brain growth but a similarly disabling outcome, characterized by intellectual disability and ASD-like symptoms.

The results presented here suggest that a shared pathophysiological mechanism might exist for idiopathic ASD. This work implies that a number of common and rare causative genetic risk factors can converge upon common mechanisms of pathogenesis. They also illustrate that directly studying neurodevelopmental processes in patients with neuropsychiatric disorders that have heterogeneous etiologies can open inroads into diagnosis and therapy (Volkmar et al., 2009).

Experimental Procedures

Induced pluripotent stem cells (iPSC)

For each proband and first degree unaffected family members, iPSC were produced using the classical retroviral approach or by a viral-free episomal reprogramming method (Okita et al., 2011). The iPSC lines from families 1123 and 03 have been previously described (Abyzov et al., 2012), and the iPSC characterization for families 07 and 1120 is shown in Figure S1 and Figure S2.

Fibroblasts and iPSC lines have been submitted to the NIMH Center for Collaborative Genomic Disorders on Mental Disorders (http://nimhgenetics.org) at the Rutgers University Cell and DNA Repository (RUCDR).

Neuronal Differentiation

We differentiated iPSC lines into telencephalic neurons using a modification of our free-floating tridimensional (3D) culture method (Mariani et al., 2012). Briefly, floating aggregates composed of manually isolated neural rosettes, which are early neural progenitors, were kept in suspension for one week under growth-promoting conditions and then for four to five weeks under conditions favoring terminal differentiation (see **Extended Experimental Procedures** and schematic outline in Figure S3A).

Electrophysiology

Conventional whole-cell patch-clamp recordings were made from individual iPSC-derived neurons with an EPC9 amplifier and PatchMaster software (HEKA). Preliminary experiments were conducted on dissociated monolayer cultures and showed that voltage-activated currents were present as early as 32 DIV and expression was stable out to 75 DIV. All subsequent characterization was performed on cells along the edges of the organoid preparations. Voltage- and current-clamp protocols were generated by the software used for data acquisition. Current and action potential amplitudes were measured in PatchMaster. Curve fitting and additional analysis was performed after exporting the data to Igor (Wavemetrics).

Transcriptome Analysis by RNAseq

We analyzed the transcriptome of 45 organoid samples, corresponding to multiple clones and three differentiation time points for 4 families, including a father and a probands. Reads

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

were aligned to the human hg19 reference genome with Tophat (Trapnell et al., 2010). Gene expression levels were estimated both as RPKM and counts using GencodeV7 gene annotation. RPKM values were estimated by using RSEQTools (Habegger et al., 2011). Counts were estimated by using BEDTools (Quinlan and Hall, 2010). After filtering out low expression genes, we inferred differentially expressed genes by edgeR (Robinson et al., 2010). Coexpressed gene modules were generated by WGCNA (Langfelder and Horvath, 2008) using $\log_2$(RPKM+1) as input, after filtering out low variance genes. DAVID (Dennis et al., 2003) and MSigDB v4.0 (Subramanian et al., 2005) were used for overrepresentation in Gene Ontologies and Canonical Pathways.

We compared the transcriptome of each of our 45 organoid samples to each of the 524 RNAseq sampels of developmental transcriptome of the human brain from the BrainSpan Atlas (www.brainspan.org), to classify our samples as corresponding to a particular brain region and developmental age. We computed the Spearman correlation matrix between the log transformed expression levels of our samples and each of the BrainSpan sampes. Each iPSC-derived neuronal sample was classified as corresponding to a particular brain sample if its correlation coefficient was within the 95% confidence interval of the maximum correlation coefficient. Average maximum correlation value is 0.86.

The RNA sequencing data are available from Gene Expression Omnibus under accession http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?token=gpkbyimupvexhwt&acc=GSE61476

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

We acknowledge support from the NIH and from the Harris Professorship fund. We also acknowledge the support team of the Yale University Biomedical High Performance Computing Center (in particular, Robert Bjornson and Nicholas Carriero). We thank Anita Huttner for culturing fibrobasts, Ying Zhang for preparing the sequencing libraries, and Nathaniel Calixto, Anahita Amiri for help in iPSC maintenance and characterization. We thank Elizabeth Jonas for facilities and help with the preliminary patch clamp experiments on dissociated cultures. We acknowledge the following grant support: NIMH MH089176, MH087879 (FMV), U54 MH066494 (Project 2 KC), and the State of Connecticut (FMV). This work was supported in part by grants from the Simons Foundation (SFARI #137055, FMV and SFARI #206929 R10981, KAP). We are grateful to all of the families at the participating Simons Simplex Collection (SSC) sites, as well as the principal investigators (A. Beaudet, R. Bernier, J. Constantino, E. Cook, E. Fombonne, D. Geschwind, R. Goin-Kochel, E. Hanson, D. Grice, A. Klin, D. Ledbetter, C. Lord, C. Martin, D. Martin, R. Maxim, J. Miles, O. Ousley, K. Pelphrey, B. Peterson, J. Piggot, C. Saulnier, M. State, W. Stone, J. Sutcliffe, C. Walsh, Z. Warren, E. Wijsman). We appreciate obtaining access to phenotypic data on SFARI Base. Approved researchers can obtain SSC population data by applying at https://base.sfari.org. We acknowledge the Yale Center for Clinical Investigation for clinical support in obtaining the biopsy specimens. We thank Dr. John Overton, the Yale Center for Genome Analysis and the Stanford Genomic Facility for advice in carrying out DNA and RNA sequencing.

References

Abrahams BS, Arking DE, Campbell DB, Mefford HC, Morrow EM, Weiss LA, Menashe I, Wadkins T, Banerjee-Basu S, Packer A. SFARI Gene 2.0: a community-driven knowledgebase for the autism spectrum disorders (ASDs). Molecular autism. 2013; 4:36. [PubMed: 24090431]

Abyzov A, Mariani J, Palejev D, Zhang Y, Haney MS, Tomasini L, Ferrandino AF, Rosenberg Belmaker LA, Szekely A, Wilson M, et al. Somatic copy number mosaicism in human skin revealed by induced pluripotent stem cells. Nature. 2012; 492:438–442. [PubMed: 23160490]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Abyzov A, Urban AE, Snyder M, Gerstein M. CNVnator: An approach to discover, genotype and characterize typical and atypical CNVs from family and population genome sequencing. Genome Res. 2011

Ariani F, Hayek G, Rondinella D, Artuso R, Mencarelli MA, Spanhol-Rosseto A, Pollazzon M, Buoni S, Spiga O, Ricciardi S, et al. FOXG1 is responsible for the congenital variant of Rett syndrome. Am J Hum Genet. 2008; 83:89–93. [PubMed: 18571142]

Bahi-Buisson N, Nectoux J, Girard B, Van Esch H, De Ravel T, Boddaert N, Plouin P, Rio M, Fichou Y, Chelly J, et al. Revisiting the phenotype associated with FOXG1 mutations: two novel cases of congenital Rett variant. Neurogenetics. 2010; 11:241–249. [PubMed: 19806373]

Casanova MF, Buxhoeveden D, Gomez J. Disruption in the inhibitory architecture of the cell minicolumn: implications for autisim. The Neuroscientist: a review journal bringing neurobiology, neurology and psychiatry. 2003; 9:496–507.

Casanova MF, Buxhoeveden DP, Switala AE, Roy E. Minicolumnar pathology in autism. Neurology. 2002; 58:428–432. [PubMed: 11839843]

Catterall WA, Goldin AL, Waxman SG. International Union of Pharmacology. XLVII. Nomenclature and structure-function relationships of voltage-gated sodium channels. Pharmacological reviews. 2005; 57:397–409. [PubMed: 16382098]

Chaste P, Klei L, Sanders SJ, Murtha MT, Hus V, Lowe JK, Willsey AJ, Moreno-De-Luca D, Yu TW, Fombonne E, et al. Adjusting head circumference for covariates in autism: clinical correlates of a highly heritable continuous trait. Biol Psychiatry. 2013; 74:576–584. [PubMed: 23746936]

Chawarska K, Campbell D, Chen L, Shic F, Klin A, Chang J. Early generalized overgrowth in boys with autism. Arch Gen Psychiatry. 2011; 68:1021–1031. [PubMed: 21969460]

Cheah CS, Yu FH, Westenbroek RE, Kalume FK, Oakley JC, Potter GB, Rubenstein JL, Catterall WA. Specific deletion of NaV1.1 sodium channels in inhibitory interneurons causes seizures and premature death in a mouse model of Dravet syndrome. Proc Natl Acad Sci U S A. 2012; 109:14646–14651. [PubMed: 22908258]

Courchesne E, Karns CM, Davis HR, Ziccardi R, Carper RA, Tigue ZD, Chisum HJ, Moses P, Pierce K, Lord C, et al. Unusual brain growth patterns in early life in patients with autistic disorder: an MRI study. Neurology. 2001; 57:245–254. [PubMed: 11468308]

Courchesne E, Mouton PR, Calhoun ME, Semendeferi K, Ahrens-Barbeau C, Hallet MJ, Barnes CC, Pierce K. Neuron number and size in prefrontal cortex of children with autism. JAMA. 2011; 306:2001–2010. [PubMed: 22068992]

Courchesne E, Pierce K, Schumann CM, Redcay E, Buckwalter JA, Kennedy DP, Morgan J. Mapping early brain development in autism. Neuron. 2007; 56:399–413. [PubMed: 17964254]

Dennis G Jr, Sherman BT, Hosack DA, Yang J, Gao W, Lane HC, Lempicki RA. DAVID: Database for Annotation, Visualization, and Integrated Discovery. Genome Biol. 2003; 4:P3. [PubMed: 12734009]

Dolen G, Osterweil E, Rao BS, Smith GB, Auerbach BD, Chattarji S, Bear MF. Correction of fragile X syndrome in mice. Neuron. 2007; 56:955–962. [PubMed: 18093519]

Eiraku M, Watanabe K, Matsuo-Takasaki M, Kawada M, Yonemura S, Matsumura M, Wataya T, Nishiyama A, Muguruma K, Sasai Y. Self-organized formation of polarized cortical tissues from ESCs and its active manipulation by extrinsic signals. Cell Stem Cell. 2008; 3:519–532. [PubMed: 18983967]

Etherton M, Foldy C, Sharma M, Tabuchi K, Liu X, Shamloo M, Malenka RC, Sudhof TC. Autism-linked neuroligin-3 R451C mutation differentially alters hippocampal and cortical synaptic function. Proc Natl Acad Sci U S A. 2011; 108:13764–13769. [PubMed: 21808020]

Fasano CA, Phoenix TN, Kokovay E, Lowry N, Elkabetz Y, Dimos JT, Lemischka IR, Studer L, Temple S. Bmi-1 cooperates with Foxg1 to maintain neural stem cell self-renewal in the forebrain. Genes Dev. 2009; 23:561–574. [PubMed: 19270157]

Habegger L, Sboner A, Gianoulis TA, Rozowsky J, Agarwal A, Snyder M, Gerstein M. RSEQtools: a modular framework to analyze RNA-Seq data using compact, anonymized data summaries. Bioinformatics. 2011; 27:281–283. [PubMed: 21134889]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Han S, Tai C, Westenbroek RE, Yu FH, Cheah CS, Potter GB, Rubenstein JL, Scheuer T, de la Iglesia HO, Catterall WA. Autistic-like behaviour in Scn1a+/− mice and rescue by enhanced GABA-mediated neurotransmission. Nature. 2012; 489:385–390. [PubMed: 22914087]

Hanashima C, Li SC, Shen L, Lai E, Fishell G. Foxg1 suppresses early cortical cell fate. Science. 2004; 303:56–59. [PubMed: 14704420]

Hutsler JJ, Zhang H. Increased dendritic spine densities on cortical projection neurons in autism spectrum disorders. Brain Res. 2010; 1309:83–94. [PubMed: 19896929]

Kang HJ, Kawasawa YI, Cheng F, Zhu Y, Xu X, Li M, Sousa AM, Pletikos M, Meyer KA, Sedmak G, et al. Spatio-temporal transcriptome of the human brain. Nature. 2011; 478:483–489. [PubMed: 22031440]

Kortum F, Das S, Flindt M, Morris-Rosendahl DJ, Stefanova I, Goldstein A, Horn D, Klopocki E, Kluger G, Martin P, et al. The core FOXG1 syndrome phenotype consists of postnatal microcephaly, severe mental retardation, absent language, dyskinesia, and corpus callosum hypogenesis. J Med Genet. 2011; 48:396–406. [PubMed: 21441262]

Lainhart JE, Bigler ED, Bocian M, Coon H, Dinh E, Dawson G, Deutsch CK, Dunn M, Estes A, Tager-Flusberg H, et al. Head circumference and height in autism: a study by the Collaborative Program of Excellence in Autism. Am J Med Genet A. 2006; 140:2257–2274. [PubMed: 17022081]

Lancaster MA, Renner M, Martin CA, Wenzel D, Bicknell LS, Hurles ME, Homfray T, Penninger JM, Jackson AP, Knoblich JA. Cerebral organoids model human brain development and microcephaly. Nature. 2013; 501:373–379. [PubMed: 23995685]

Langfelder P, Horvath S. WGCNA: an R package for weighted correlation network analysis. BMC bioinformatics. 2008; 9:559. [PubMed: 19114008]

Lawrence YA, Kemper TL, Bauman ML, Blatt GJ. Parvalbumin-, calbindin-, and calretinin-immunoreactive hippocampal interneuron density in autism. Acta neurologica Scandinavica. 2010; 121:99–108. [PubMed: 19719810]

Mariani J, Simonini MV, Palejev D, Tomasini L, Coppola G, Szekely AM, Horvath TL, Vaccarino FM. Modeling human cortical development in vitro using induced pluripotent stem cells. Proc Natl Acad Sci U S A. 2012; 109:12770–12775. [PubMed: 22761314]

Martynoga B, Morrison H, Price DJ, Mason JO. Foxg1 is required for specification of ventral telencephalon and region-specific regulation of dorsal telencephalic precursor proliferation and apoptosis. Dev Biol. 2005; 283:113–127. [PubMed: 15893304]

Mencarelli MA, Spanhol-Rosseto A, Artuso R, Rondinella D, De Filippis R, Bahi-Buisson N, Nectoux J, Rubinsztajn R, Bienvenu T, Moncla A, et al. Novel FOXG1 mutations associated with the congenital variant of Rett syndrome. J Med Genet. 2010; 47:49–53. [PubMed: 19578037]

Ogiwara I, Miyamoto H, Morita N, Atapour N, Mazaki E, Inoue I, Takeuchi T, Itohara S, Yanagawa Y, Obata K, et al. Nav1.1 localizes to axons of parvalbumin-positive inhibitory interneurons: a circuit basis for epileptic seizures in mice carrying an Scn1a gene mutation. J Neurosci. 2007; 27:5903–5914. [PubMed: 17537961]

Okita K, Matsumura Y, Sato Y, Okada A, Morizane A, Okamoto S, Hong H, Nakagawa M, Tanabe K, Tezuka K, et al. A more efficient method to generate integration-free human iPS cells. Nat Methods. 2011; 8:409–412. [PubMed: 21460823]

Parikshak NN, Luo R, Zhang A, Won H, Lowe JK, Chandran V, Horvath S, Geschwind DH. Integrative functional genomic analyses implicate specific molecular pathways and circuits in autism. Cell. 2013; 155:1008–1021. [PubMed: 24267887]

Piven J, Arndt S, Bailey J, Andreasen N. Regional brain enlargement in autism: a magnetic resonance imaging study. J Am Acad Child Adolesc Psychiatry. 1996; 35:530–536. [PubMed: 8919716]

Pizzarelli R, Cherubini E. Developmental regulation of GABAergic signalling in the hippocampus of neuroligin 3 R451C knock-in mice: an animal model of Autism. Frontiers in cellular neuroscience. 2013; 7:85. [PubMed: 23761734]

Quinlan AR, Hall IM. BEDTools: a flexible suite of utilities for comparing genomic features. Bioinformatics. 2010; 26:841–842. [PubMed: 20110278]

Robinson MD, McCarthy DJ, Smyth GK. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics. 2010; 26:139–140. [PubMed: 19910308]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Rubenstein JL. Three hypotheses for developmental defects that may underlie some forms of autism spectrum disorder. Current opinion in neurology. 2010; 23:118–123. [PubMed: 20087182]

Shoichet SA, Kunde SA, Viertel P, Schell-Apacik C, von Voss H, Tommerup N, Ropers HH, Kalscheuer VM. Haploinsufficiency of novel FOXG1B variants in a patient with severe mental retardation, brain malformations and microcephaly. Human genetics. 2005; 117:536–544. [PubMed: 16133170]

Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, Paulovich A, Pomeroy SL, Golub TR, Lander ES, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proc Natl Acad Sci U S A. 2005; 102:15545–15550. [PubMed: 16199517]

Sudhof TC. Neuroligins and neurexins link synaptic function to cognitive disease. Nature. 2008; 455:903–911. [PubMed: 18923512]

Tabuchi K, Blundell J, Etherton MR, Hammer RE, Liu X, Powell CM, Sudhof TC. A neuroligin-3 mutation implicated in autism increases inhibitory synaptic transmission in mice. Science. 2007; 318:71–76. [PubMed: 17823315]

Tang G, Gudsnuk K, Kuo SH, Cotrina ML, Rosoklija G, Sosunov A, Sonders MS, Kanter E, Castagna C, Yamamoto A, et al. Loss of mTOR-Dependent Macroautophagy Causes Autistic-like Synaptic Pruning Deficits. Neuron. 2014

Trapnell C, Williams BA, Pertea G, Mortazavi A, Kwan G, van Baren MJ, Salzberg SL, Wold BJ, Pachter L. Transcript assembly and quantification by RNA-Seq reveals unannotated transcripts and isoform switching during cell differentiation. Nat Biotechnol. 2010; 28:511–515. [PubMed: 20436464]

Vaccarino FM, Grigorenko EL, Smith KM, Stevens HE. Regulation of cerebral cortical size and neuron number by fibroblast growth factors: implications for autism. J Autism Dev Disord. 2009; 39:511–520. [PubMed: 18850329]

Volkmar FR, State M, Klin A. Autism and autism spectrum disorders: diagnostic issues for the coming decade. Journal of child psychology and psychiatry, and allied disciplines. 2009; 50:108–115.

Willsey AJ, Sanders SJ, Li M, Dong S, Tebbenkamp AT, Muhle RA, Reilly SK, Lin L, Fertuzinhos S, Miller JA, et al. Coexpression networks implicate human midfetal deep cortical projection neurons in the pathogenesis of autism. Cell. 2013; 155:997–1007. [PubMed: 24267886]

Xuan S, Baptista CA, Balas G, Tao W, Soares VC, Lai E. Winged helix transcription factor BF-1 is essential for the development of the cerebral hemispheres. Neuron. 1995; 14:1141–1152. [PubMed: 7605629]

Yu FH, Mantegazza M, Westenbroek RE, Robbins CA, Kalume F, Burton KA, Spain WJ, McKnight GS, Scheuer T, Catterall WA. Reduced sodium current in GABAergic interneurons in a mouse model of severe myoclonic epilepsy in infancy. Nat Neurosci. 2006; 9:1142–1149. [PubMed: 16921370]

Zoghbi HY, Bear MF. Synaptic dysfunction in neurodevelopmental disorders associated with autism and intellectual disabilities. Cold Spring Harbor perspectives in biology. 2012; 4

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Highlights

- iPSC-derived telencephalic organoids reflect human midfetal telencephalic development
- Inhibitory neurons are overproduced in organoids from patients with idiopathic autism
- Overproduction of inhibitory neurons is caused by increased FOXG1 gene expression

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

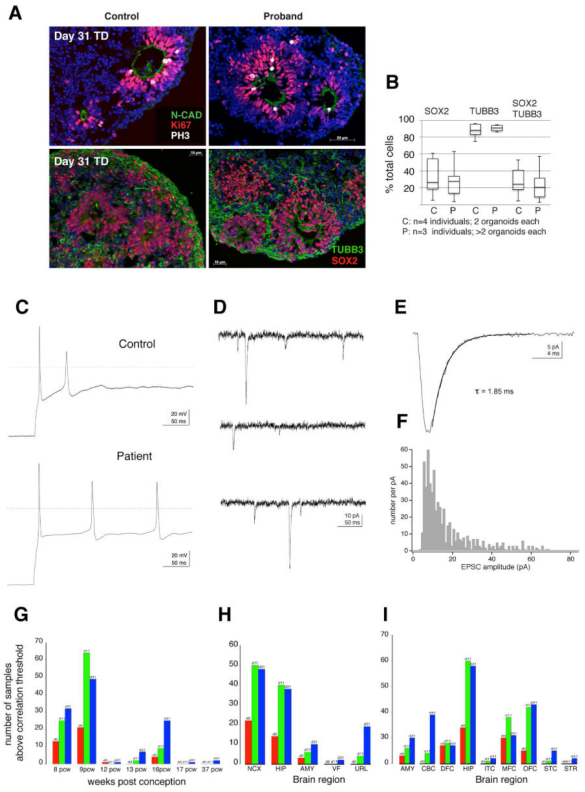


Figure 1. ASD organoids display normal neuronal differentiation potential

A, Both control-derived and ASD-derived organoids express markers for proliferating neural progenitors (SOX2), and the neuronal markers TUBB3. The organoids have apico-basal polarity with *N*-CADHERIN$^+$ apical end feet of radial glial cells and pH3$^+$ cells undergoing mitosis at the apical side of the neuroepithelium. **B**, Stereological quantification of SOX2$^+$ proliferating radial glia progenitors and TUBB3$^+$ neuronal cells at TD31. Scale bars, 10 μm or 20 μm as indicated. **C**, Action potentials elicited by depolarizing current injections in iPSC-derived neurons from a patient and the corresponding parental control. The dashed lines indicate a membrane potential of 0 mV. **D**, Examples of spontaneous excitatory postsynaptic currents (EPSCs) recorded at a holding potential of −70 mV in a neuron from a parental control that was maintained in vitro for 52 days. **E**, average EPSC obtained from 30 spontaneous events. The decay of the current was fitted with a single exponential that gave a time constant of 1.85 ms. **F**, Histogram of the amplitude of spontaneous inward EPSCs recorded from the same neuron (604 events). The overall frequency of events detected in this neuron was 0.81 per second. **G–I,** Transcriptome correlation between organoids at terminal differentiation day 0 (d0) (rosette stage), day 11 and 31 (d11 and d31) (n= 45 samples, see Table S1) and postmorterm human brain samples from the BRAINSPAN project (n=524 samples). The X-axis shows the postconceptional age in weeks or the brain region of the BrainSpan postmorterm brain samples. The Y-axis is the number of times each iPSC-derived neuron sample was classified. Classification was based on the maximum correlation coefficient and on the 95% confidence interval of the maximum correlation (see Supplementary Methods). AMY, amygdala; CBC, cerebellar cortex; DFC, dorsolateral frontal cortex; HIP, hippocampus; ITC, inferolateral temporal cortex; MFC, medial

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

prefrontal cortex; OFC, orbital frontal cortex; STC, superior temporal cortex; STR, striatum. In **I**, X-axis shows BRAINSPAN agglomerated brain regions (i.e. more brain regions were merged into a single 'larger' region): NCX (i.e. FC, PC, TC, OC), neocortex; HIP, hippocampus; AMY, amygdala; VF (i.e. VF, MGE, LGE, CGE, STR), ventral forebrain; URL (i.e. CBC, CB, URL), upper rhombic lip. See also Figure S3.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

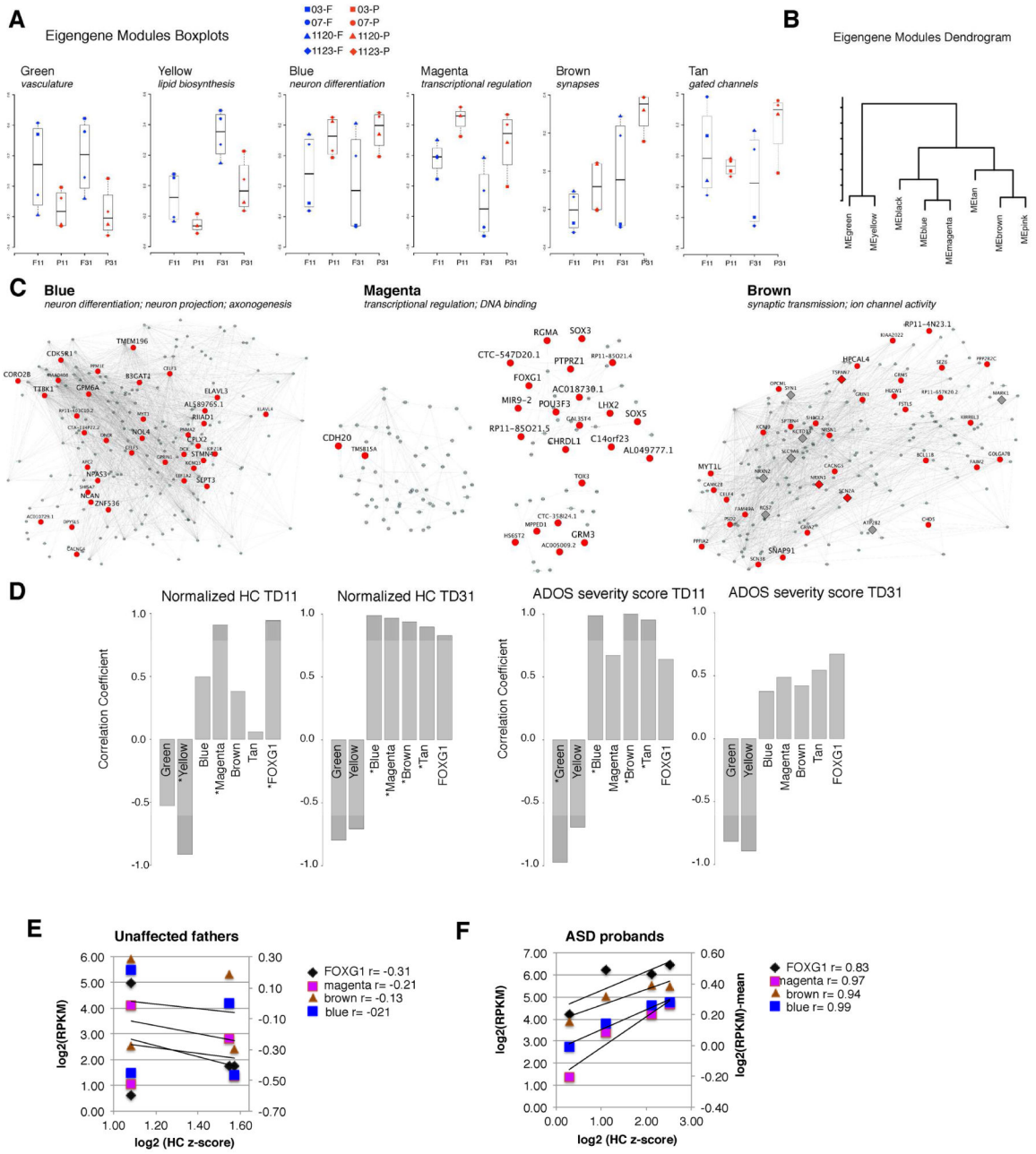


Figure 2. WGCNA network in neuronal cells from ASD patients and unaffected family controls and correlation with clinical phenotypes

A, Relationship between the modules' eigengenes with diagnosis and time in culture. TD11: 11 days of terminal differentiation; TD31: 31 days of terminal differentiation; F: fathers; P: probands. **B**, Module-to-module relationship by hierarchical clustering. **C**, Top 200 hub genes networks for three representative modules. Circles: genes; diamonds: genes overlapping with genes in the SFARI database classified as associated to ASD; red: genes overexpressed at either TD11 or TD31; grey: no changes in gene expression; larger font: genes differentially expressed at both TD11 and TD31; **D**, Pearson's correlation between modules' eigengenes with log transformed HC Z-scores and ADOS severity scores at TD11

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

and TD31; *nominal p-value <0.1; **E**, **F**, Pearson's correlation between HC Z-scores and modules' eigengenes or FOXG1 expression levels for the unaffected fathers (**E**) and the affected probands (**F**) at TD31. See also Figure S4.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

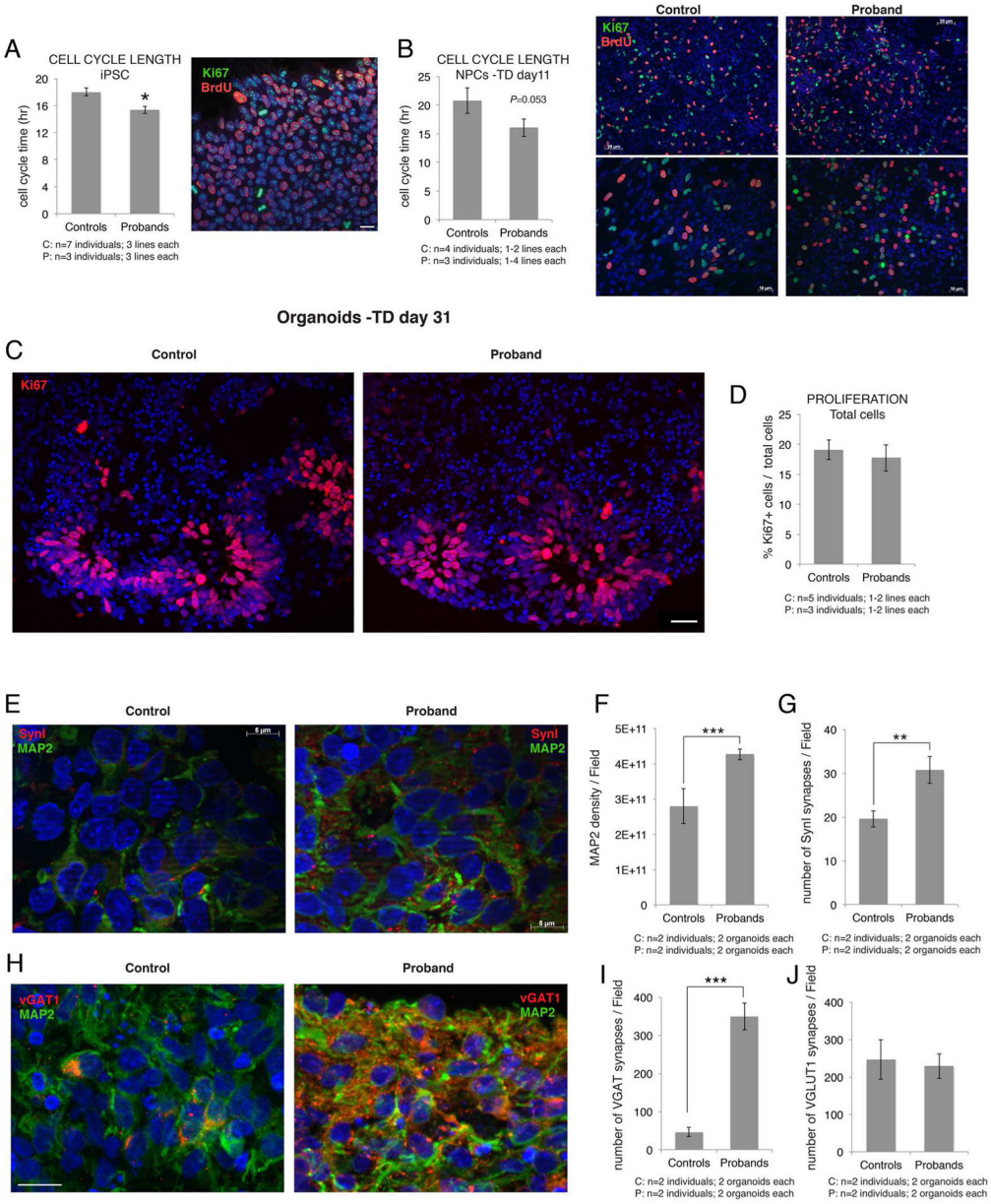


Figure 3. Decreased cell cycle length and increased neuronal/synaptic differentiation in ASD probands

A, B, Cell cycle time determined by the formula Tc= Ts/(%BrdU/Ki67), where Tc= cell cycle time and Ts= S phase time. **A**, Representative image of double immunostaining for BrdU (red) and Ki67 (green) of undifferentiated iPSCs from a control individual and (**B**) early neuronal progenitors in monolayer cultures at TD11. *$p < 0.05$, ANOVA with family as covariant. To prepare monolayer cultures for neuronal progenitors the neural rosettes were dissociated into single cells and cultured in adhesion as monolayers until TD11. **C, D**, Representative images (**C**) and stereological quantification (**D**) of the proportion of proliferating Ki67^{+} neuronal progenitor cells in both control and ASD-derived organoids at TD31. **E-J**, Increased neuronal maturation and synaptic formation in ASD-derived neurons

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

(TD 31). Representative images of controls and probands organoids at TD 31 labeled with MAP2 and SynI (**E**), or MAP2 and VGAT (**H**), and relative quantification of MAP2 density (**F**), number of $SynI^+$ puncta (**G**), number of $VGAT^+$ puncta (**I**), and number of $VGLUT1^+$ puncta (**J**). The data in (**F**, **G**, **I**, **J**) are presented as means ± s.e.m; $^{**}p < 0.01$, $^{***}p < 0.001$; t test analysis. Scale bars, 10 μm (**A**), 10–20 μm (**B**, as indicated), 10 μm (**C**), 5 μm (**E**, **H**).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

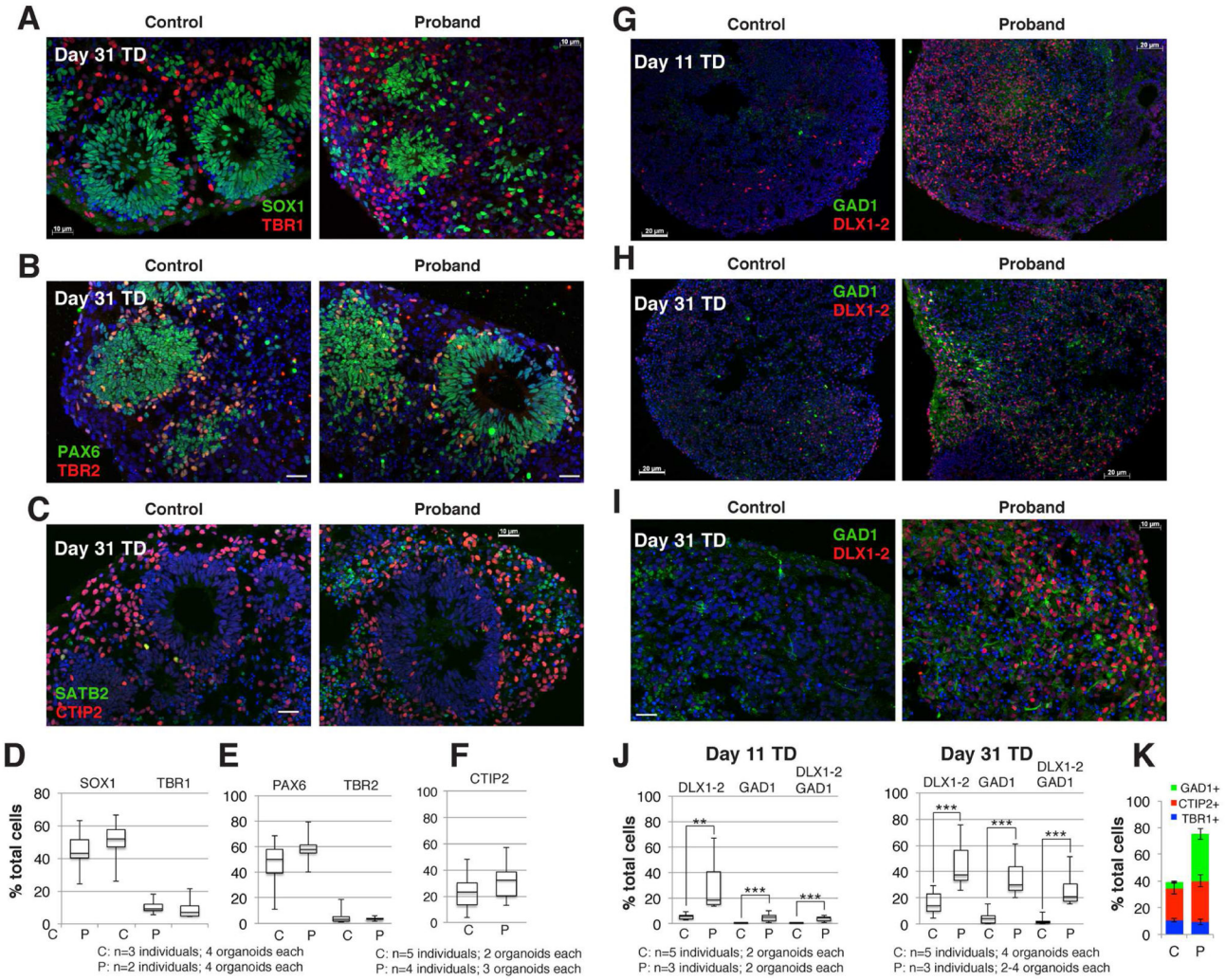


Figure 4. ASD organoids show imbalance between glutamatergic and GABAergic neuron fate

Representative images of control-derived and ASD proband-derived organoids: **A**-**F**, Immunostaining and respective stereological quantification of SOX1$^+$ and PAX6+ proliferating radial glia progenitors, cortical excitatory TBR2$^+$ intermediate progenitors, and more mature TBR1$^+$ and CTIP2$^+$ excitatory neurons at TD31. **G**-**J**, Immunostaining and quantification of GABAergic inhibitory progenitor cells (DLX1-2$^+$) and mature GABAergic interneurons (GAD1$^+$) at TD11 and TD31. **K**, Box plots showing percentages of inhibitory (GAD1$^+$) and excitatory (TBR1$^+$, CITIP2$^+$) neurons in ASD-derived and control-derived organoids at TD31. Total cells are estimated by counting DAPI$^+$ nuclei. C=Controls, P=Probands. $^{**}p < 0.01$, $^{***}p < 0.001$, t test analysis. Scale bars, 10 μm (**A**, **C**, **E**, **G**, **K**), 20 μm (**I**, **J**). See also Figure S4 and Figure S5.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

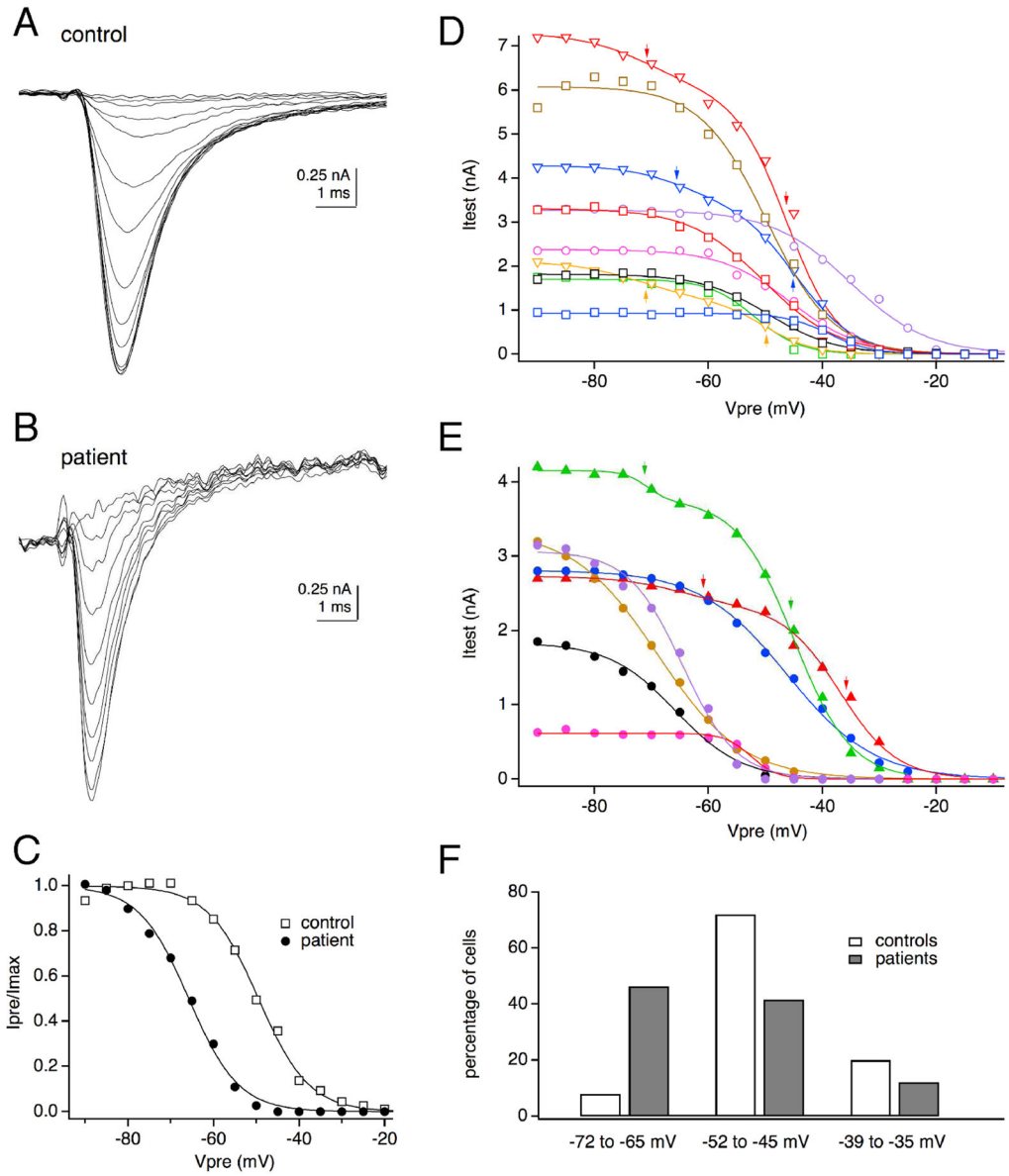


Figure 5. Sodium currents in iPSC-derived neurons from ASD patients and controls show different voltage dependence of steady-state inactivation

A, **B**, Inward sodium currents evoked by depolarizing test jumps to −20 mV from pre-test potentials of −90 mV to −25 mV (**A**) or −45 mV (**B**) in steps of 5 mV. **C**, Peak inward current amplitudes as a function of pre-test potential from the data in **A** and **B**. The results were normalized to Imax values obtained for each neuron from fits to the raw data (as in **D** and **E**). **D**, **E**, Steady-state inactivation data obtained from 10 control neurons (**D**) and seven patient neurons (**E**). Data for individual cells are shown with different symbols and colors (normalized data in **C** are shown here with the same symbols: control, open black square; patient, filled black circle). For some neurons, the results were better fitted as the sum of two components (arrowheads point to $E_{h1/2}$ values for each component). **F**, Bar graph depicting the percentage of control and patient neurons (n = 10 and 7) with $E_{h1/2}$ values that fell

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

within the indicated ranges. When two components were present, the fractional amplitudes of each were used in the calculation of mean percentages for each group. Cells that gave $E_{h1/2}$ values ≤ −65 mV gave half-activation voltages that ranged from −39.1 to −55.8 mV, a phenotype most consistent with the $Na_v1.1$ brain sodium channel isoform (Catterall et al., 2005).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

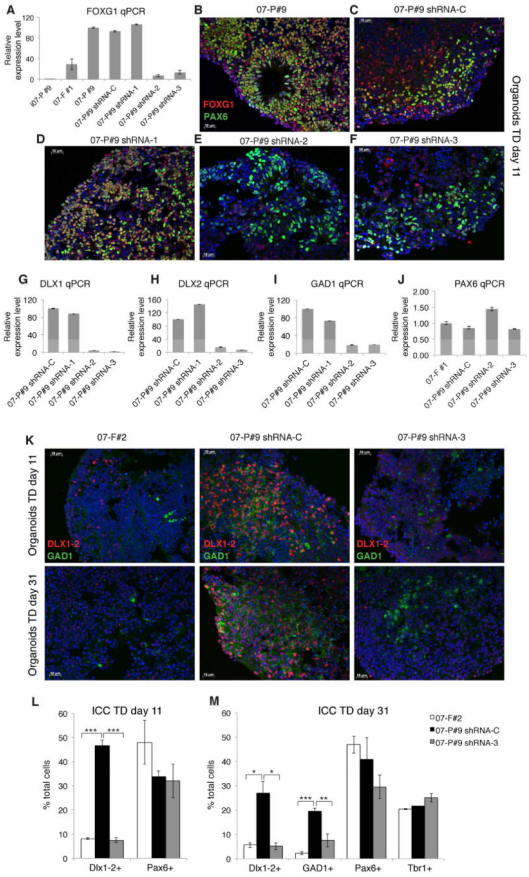


Figure 6. FOXG1 knockdown in ASD-derived organoids is able to restore the balance between GABA/glutamate neuron fate

A, Relative expression levels of FOXG1 by qPCR among non-virally transduced undifferentiated iPSCs from proband #9 (i07-P#9), TD11 organoids from the proband (07-P#9), his father (lines 07-F#1), and proband's organoids harboring a non-targeting shRNA-control (shRNA-C) or three different shRNAs targeting FOXG1 (shRNA-1, shRNA-2, shRNA-3). **B**–**F**, Double immunostaining for FOXG1 and PAX6 in TD11 non-transduced proband's organoids (**B**), or transduced with shRNA-C (**C**), shRNA-1 (**D**), shRNA-2 (**E**), shRNA-3 (**F**). **G**–**J**, qPCR for DLX1 (**G**), DLX2 (H), GAD1 (**I**), and PAX6 (**J**) in TD11 organoids from shRNA-C and shRNA-1/2/3. **K**, DLX1-2 and GAD1 double immunostaining in organoids derived from the father or the proband transduced with shRNA-C or shRNA-3 at TD11 and TD 30. **L**, **M**, Stereological quantification of immunocytochemical (ICC) staining for GABAergic (DLX1-2, GAD1) and glutamatergic markers (PAX6, TBR1) at TD11 (**L**) and TD 31 (**M**). (Sample names: 07= family name from which iPSCs were derived; F=Father; P=Proband; #=iPS clone number). Data in (**G**–**J**, **L**, **M**) are presented as means ± s.e.m; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, t test analysis. Scale bars, 10 μm.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

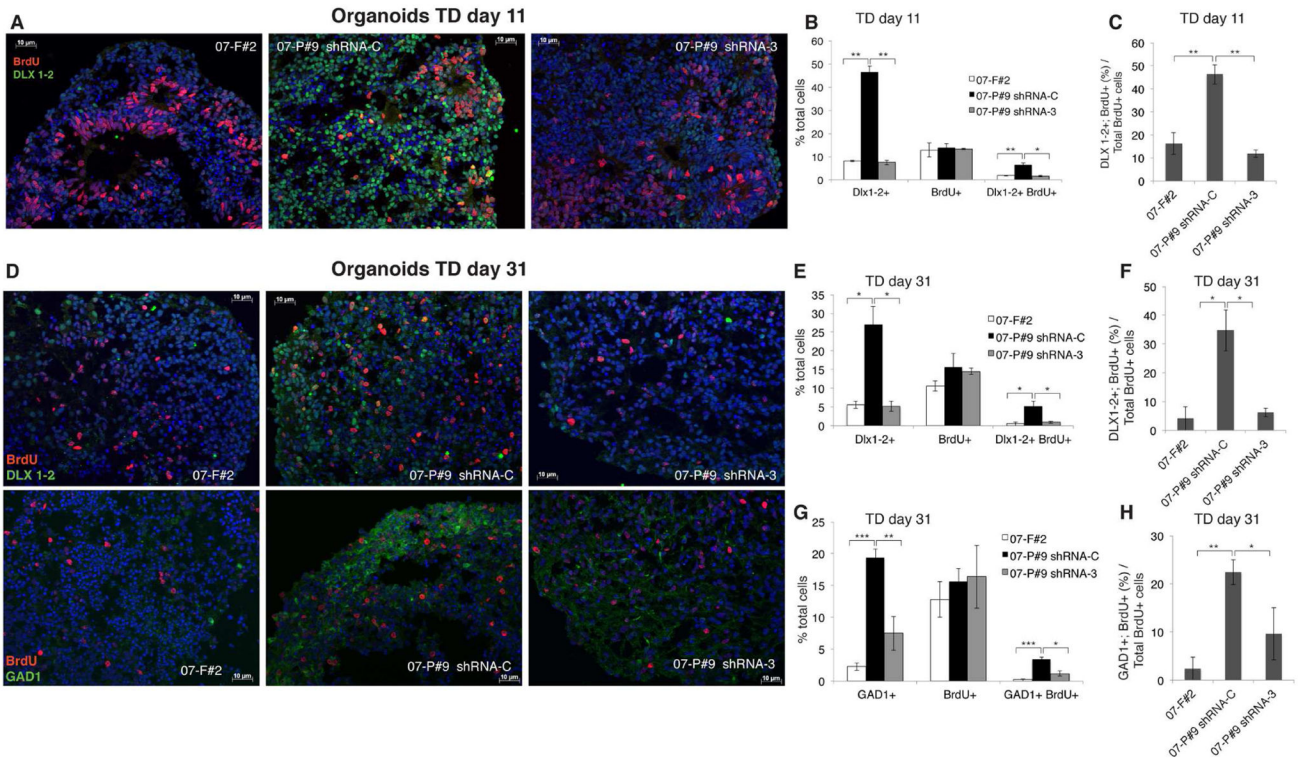


Figure 7. Increased proportion of proliferating GABAergic neuronal progenitors and mature GABAergic interneurons at TD31 in ASD-derived organoids

Representative images (**A**, **D**) and stereological quantification of BrdU$^+$/DLX1-2$^+$ proliferating cells (**B**, **C**, **E**, **F**) and BrdU$^+$/GAD1$^+$ neurons (**G**, **H**) in TD11 and TD31 organoids derived from 07-F#2 (father), 07-P#9 shRNA-C, and 07-P#9 shRNA-3 (proband's iPSC lines transduced with shRNA-C or with shRNA-3). The selectively increased proportion of DLX 1-2$^+$/BrdU$^+$ double positive cells in patient-derived organoids at both TD11 (**B**, **C**) and TD31 (**E**, **F**), was restored to a physiological level after FOXG1-knockdown at both time points. The increased proportion of proliferating DLX1-2 GABAergic progenitors (**A** and **D**, upper panel) resulted in an overproduction of more mature GABAergic GAD1+/BrdU+ double positive interneurons (**D**, bottom panel), overproduction that was restored to physiological levels after FOXG1-knockdown (**G**,**H**). Data in (**B**, **C**, **E**, **F**, **G**, **H**) are presented as means ± s.e.m; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, t test analysis. Scale bars, 10 μm. See also Figures S6, S7.